Gungnir: Autoregressive Model for Unified Generation of IPv6 Fully Responsive Prefixes

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***Abstract*—With the widespread adoption of IPv6, its vast address space presents significant challenges for network asset discovery. Traditional exhaustive scanning approaches are no longer practical, while strategies based on target generation algorithms are increasingly undermined by the existence of Fully Responsive Prefixes (FRPs)—prefixes in which all addresses appear responsive to probing. FRPs distort scanning results, introducing bias and inefficiency. Existing FRP probing techniques suffer from limited scalability, poor accuracy, and restricted applicability in large-scale IPv6 environments.**

To this end, we propose *Gungnir*, a multi-protocol unified FRP probing algorithm based on autoregressive semantic modeling. *Gungnir* captures the intricate relationships between FRP patterns and their influencing factors through a deep semantic learning architecture. It leverages prefix inference and a granularity correction mechanism to accurately predict and validate FRPs, while mitigating errors from incorrect prefix-length estimation. Extensive experiments demonstrate that *Gungnir* outperforms state-of-the-art techniques, achieving up to 27× higher efficiency, 4.2× wider address space coverage, and broader coverage of both autonomous systems and routing prefixes under the same probing budget. Beyond performance, we further analyze the service and port distributions of the discovered FRPs, uncovering operational patterns and potential security implications. These insights offer valuable guidance for IPv6 measurement, address discovery, and network defense.

I. Introduction

Internet-wide network scanning, as the core technology of cyberspace resource mapping, obtains the response characteristics of the target network by actively sending customized probe packets. For example, tools such as ZMap [1] enable high-performance active address scanning. It provides basic data support for researching Internet infrastructure security [2], [3], conducting geolocation analysis [4], and analyzing the impact of the Internet on the real world [5], and plays a crucial role in the field of network measurement [6]–[10]. However, with the large-scale deployment of the IPv6 protocol [11], the huge space brought by its 128-bit address makes network scanning face new challenges. Within this space lies a special class of prefixes—Fully Responsive Prefixes (FRPs)—which span large ranges where every address appears responsive to probing [12], [13]. For example, if all addresses within a target prefix respond to ICMPv6 Echo messages, the prefix can be classified as an FRP for ICMPv6. Nevertheless, it is evident that such responses cannot originate from a single host.

FRPs have significantly influenced the widely used target generation algorithms (TGAs) [14]–[22] employed in active address probing—one of the current research hotspots. For example, all addresses under a /32 prefix (totaling 2^{96} addresses) may appear responsive, leading scanners to treat them as legitimate targets. This proliferation of non-authentic or synthetic addresses introduces substantial noise into scanning results, potentially causing misinterpretations—such as assuming that multiple addresses within an FRP correspond to distinct services hosted on separate machines [12], [23]. The widespread deployment of FRPs also reflects various intentional uses: running multiple services on a single server, enabling CDN providers to implement server-side load balancing, or facilitating security researchers to deploy honeypots for traffic analysis [12], [23]–[25]. As a result, understanding the global distribution and behavior of FRPs has become increasingly critical for the development of next-generation scanning methodologies. Accurate identification and handling of FRPs not only improve the fidelity of scanning outcomes but also lay the groundwork for their constructive application in network research and operations.

Previous research efforts [26] have identified the presence of such prefixes in active address probing. To mitigate their impact, a method known as MAPD [12] was proposed for identifying these prefixes. However, MAPD fundamentally relies on an exhaustive generation approach that produces a set of random candidate addresses for probing—a strategy that becomes impractical given the vastness of the IPv6 address space. Furthermore, MAPD is limited to detecting seed addresses (i.e., known active IPv6 addresses), rendering it ineffective for comprehensive, Internet-wide FRP discovery. To overcome these limitations, Luori [25] efficiently uncovers FRPs via active probing. It formulates FRP detection as a reinforcement learning problem, leveraging Monte Carlo Tree Search (MCTS) to guide probing and improve efficiency.

Despite recent advancements, FRP probing continues to face several key challenges: *(1) Enormous Decision Space.* FRP probing requires the identification of target prefixes within the vast IPv6 address space. Existing methods such as MAPD [12], which relies on exhaustive enumeration of prefixes with fixed lengths, and Luori [25], which adopts a tree-based approach for layer-by-layer expansion, struggle with accurately predicting both the lengths and patterns of FRPs. These limitations lead to inefficiencies, especially when incorrect prefix lengths are assumed. For instance, mistakenly treating a /64 prefix as a /80 can result in significant resource waste and introduce substantial noise into the dataset due to unnecessary probing. *(2) Multi-Protocol and Port-Type Probing Complexity.* Previous work [12], [25] also falls short

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in addressing FRP probing across the full range of protocol and port types. Our study reveals that FRPs vary significantly in behavior depending on protocol (e.g., TCP, UDP) and port combinations, with distinct activity states. Effective discovery thus requires Internet-wide probing under each protocol-port pair, which introduces an enormous overhead. For example, considering both TCP and UDP, there are 65,536×2=131,072 possible port-type combinations to account for. Consequently, a highly scalable algorithm is needed to support comprehensive FRP probing across all protocols and ports in the IPv6 space, while maintaining efficiency and accuracy.

To this end, we propose *Gungnir*, a novel FRP probing algorithm that unifies support for multiple protocols and port types through an autoregressive modeling framework. The core idea behind *Gungnir* is to leverage deep learning–based semantic understanding to capture complex FRP patterns. Specifically, the algorithm encodes FRP patterns along with associated contextual factors—including protocol and port types, routing prefixes, originating ASes, affiliated organizations, and business categories—and learns their intricate relationships. By incorporating a self-attention mechanism and positional encoding, *Gungnir* effectively extracts and represents the semantic dependencies within sequential data, such as correlations between business types and address patterns. During prefix inference, *Gungnir* utilizes this learned semantic information to generate a list of candidate FRPs. To mitigate the risk of "FRP Misleading" caused by granularity mismatches in prefix lengths, the algorithm employs a granularity correction mechanism, thereby enhancing accuracy. Overall, *Gungnir* enables efficient and precise probing of FRPs across the IPv6 address space.

Contributions. The primary contributions of this work are as follows:

- We propose the first FRP prediction algorithm unified across multiple protocols and ports. By leveraging an autoregressive model for the semantic learning of FRP patterns, our approach overcomes the limitations of traditional methods and enables accurate prediction of FRPs across a wide range of protocol and port types.
- We implement a prototype of *Gungnir*, and evaluation results show that under the same prefix budget, *Gungnir* achieves nearly 27× faster probing speed across various protocols and ports compared to the state-of-the-art method.
- This study reveals the intrinsic pattern of FRPs and their patterns in distribution, and deeply analyzes the statistical characteristics, pattern information of FRPs. The deployment We reveal potential security risks such as outdated versions, weak cryptographic compatibility, and shared certificates. For the community to better carry out the research work on active addresses and FRPs, we make our code and all data public, and construct a list of multi-active types of FRPs with the widest coverage to the best of our knowledge. Code is available at https://github.com/THUNAME/Gungnir.

II. Background and Related Work

A. Fully Responsive Prefixes

Before the formalization of FRPs, Murdock et al. [26] observed a notable phenomenon in IPv6 active address probing: a significant proportion of addresses within the same prefix responded uniformly to Internet-wide scans, a characteristic they termed aliased prefixes. Their active probing experiments revealed that 98% of responsive addresses resided in regions exhibiting large-scale aliasing, despite these addresses ultimately resolving to the same host or network service via FRPs configured across different protocol ports [27]. Building on this discovery, Zirngibl et al. [13] rigorously defined FRPs as prefixes demonstrating universal address responsiveness (i.e., a 100% probe reply rate), while critically differentiating them from aliased prefixes by relaxing the constraint of single-host control. This conceptual framework establishes aliased prefixes as a proper subset of FRPs—a hierarchical relationship consistently maintained in our subsequent analysis. A parallel phenomenon exists in IPv4 networks, where Sattler et al. [27] identified highly responsive prefixes (HRPs). These prefixes exhibit responsiveness from over 90% of their constituent addresses for specific ports. Unlike their IPv6 counterparts, IPv4 HRPs may encompass multiple hosts, including content delivery networks (CDNs) or TCP proxy infrastructures.

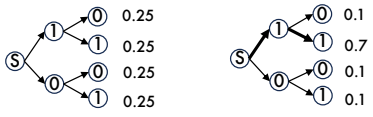
Recent research has increasingly focused on the measurement biases introduced by these phenomena. Oliver Gasser et al. [12] demonstrated that aliased prefixes artificially inflate hitlist address counts by mapping multiple entries to single devices, thereby distorting assessments of network scale and topology. Similarly, Lion Steger et al. [28] identified aliased prefixes as a critical consideration in IPv6 measurement methodologies.

B. FRP Probing

For FRP probing, earlier studies [26] employed a straightforward method to detect aliased prefixes. First, they selected the /96 prefix containing the responding probe addresses, then randomly generated three addresses within that prefix. Three TCP SYN probes were sent to each address, and if all three responded to at least one probe, the prefix was deemed aliased. However, this approach introduces bias, prompting Gasser et al. to propose Aliased Prefix Detection (APD) [12]. APD enhances reliability by randomly selecting 16 addresses within the 0-f sub-prefix of a candidate prefix. If all 16 addresses respond, the prefix is labeled as aliased. Subsequent FRP probing efforts have largely adopted the APD methodology, evolving into MAPD [14], [17], [21], [25], [29]. MAPD exhaustively enumerates all possible candidate prefixes and applies aliased prefix probing. As noted by [25], these methods do not verify whether the probed addresses resolve to the same host, effectively making them FRP probing techniques rather than strict alias detection methods.

Most recently, Luori [25] introduced an active probing framework for FRP that leverages MCTS to transform the vast IPv6 search space into a reinforcement learning decision-making process. While effective, its hierarchical tree expansion

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$H_{pattern}(S) = 2$ $H_{pattern}(S) = 1.36$

Fig. 1: Calculated values of pattern entropy under random and clustered distributions

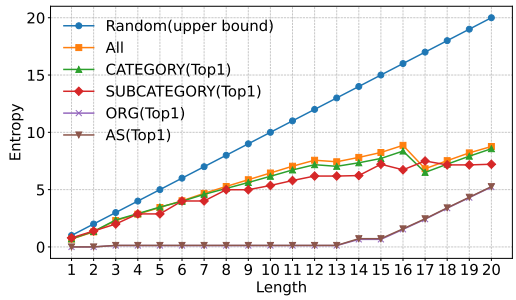


Fig. 2: Variation of pattern entropy in different types

design introduces length dependency, limiting its scalability in FRP probing.

III. Problem Analysis and Definition

The purpose of FRP probing is to identify as many FRPs as possible within a given target routing prefix, building upon the active address probing framework proposed in [30] by first analyzing the structural patterns of FRPs. FRPs consist of two main components—the fixed routing prefix (determined by BGP announcements) and variable FRP patterns—and our analysis focuses on the latter. We model these patterns as a prefix tree where the root represents the routing prefix and each subsequent layer corresponds to binary expansions of the pattern, revealing distinct distributional characteristics that we quantify through the novel concept of *pattern entropy*. This approach enables systematic exploration of FRP patterns while accounting for their hierarchical structure and variability across the prefix tree.

Proposition 1: Given a seed set S, let $P(x)$ denote the probability of reaching leaf node x in the prefix tree. The FRP pattern entropy $H_{\text{pattern}}(S)$ for a layer of length l is defined:

$$H_{pattern}(S) = -\sum P(x)\log P(x) \tag{1}$$

The pattern entropy $H_{\text{pattern}}(S)$ effectively quantifies the distribution characteristics of FRP patterns. As illustrated in Figure 1, the left side demonstrates a random distribution (maximal entropy), while the right side shows a clustered distribution (lower entropy). A smaller pattern entropy indicates a more regular and structured pattern distribution in FRPs. Mathematically, it can be proven that $H_{\text{pattern}}(S)$ reaches its maximum value when patterns are uniformly and randomly distributed across the prefix tree.

$$MAX(H_{pattern}(S)) = l, P(x_i) = \frac{1}{l} \tag{2}$$

Figure 2 presents the pattern entropy statistics for the current FRP seed set [25], categorized by AS, organization, and business category (sorted by pattern count). The horizontal axis represents pattern length l, while the vertical axis denotes pattern entropy. Notably, all observed entropy values remain significantly below the theoretical upper bound (blue line), which represents the entropy of randomly distributed patterns. This substantial gap indicates that FRP patterns exhibit strong regularity, suggesting potential for optimization through pattern mining to enhance probing accuracy.

Furthermore, the analysis reveals that segmenting FRPs based on business relationships consistently reduces pattern entropy across most dimensions. This entropy reduction implies that granularity changes—whether by length, AS, or other attributes—significantly influence FRP pattern distribution. Our findings, corroborated through rigorous data analysis, demonstrate for the first time that FRPs exhibit pronounced clustering behavior across multiple dimensions (e.g., length, AS), with complex interactions between these factors.

Based on the observed regularity in FRP distribution across organizations and business types, as well as its pronounced clustering behavior at different prefix pattern granularities, we formulate the mathematical objective for FRP probing as:

$$\min_{\theta} \mathrm{KL}\left(P_d(x) \| P_\theta(x)\right) \tag{3}$$

where d represents the dataset, θ denotes the probing algorithm parameters, and $\mathrm{KL}(\cdot)$ is the Kullback-Leibler divergence [31], which quantifies the discrepancy between the data distribution $P_d(x)$ and the algorithm's output distribution $P_\theta(x)$.

Interestingly, this objective aligns with the optimization goals of generative AI algorithms [32], [33], which have demonstrated remarkable success across diverse domains. Motivated by this convergence, we leverage generative AI principles to design *Gungnir*, our novel FRP probing framework.

IV. *Gungnir* Design

This section presents the detailed design of *Gungnir*. We begin with an overview of the approach, followed by a detailed explanation of each module's design.

A. *Overview of* Gungnir

The goal of *Gungnir* is to leverage an autoregressive model to capture the patterns of known types of FRP, enabling efficient discovery of FRPs in the wild across different protocols and ports. The FRP probing process of *Gungnir* consists of three steps: semantic learning, prefix inference, and online scanning. As shown in Figure 4, to achieve highly scalable FRP pattern prediction, *Gungnir* first employs an autoregressive model for semantic learning to capture the complex relationships between FRP patterns and their correlates (§IV-B). This probabilistic model is then used to generate semantic representations of FRP patterns. During the prediction phase, *Gungnir* leverages the learned semantic information of routing prefixes to predict potential FRP patterns and generates candidate FRPs (§IV-C). In the scanning phase, *Gungnir* performs online scanning of these candidate FRPs, collects results, and determines whether the candidates are actual FRPs based on the probing feedback. For confirmed FRPs, granularity correction is applied to derive a precise set

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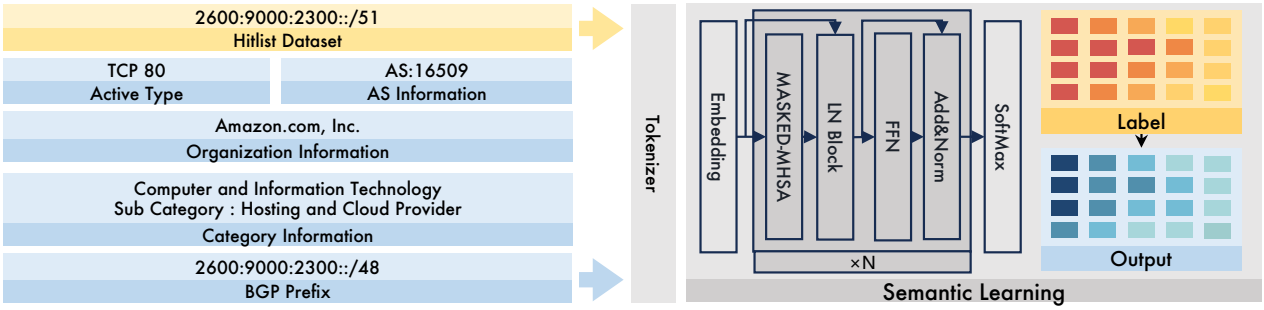


Fig. 3: Semantic learning process and model architecture diagram. The model performs semantic learning by learning multiple associative information.

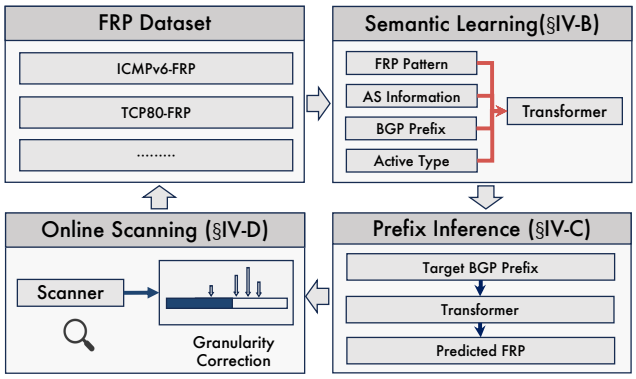


Fig. 4: Overview of *Gungnir*. The orange lines indicate the pattern information of *Gungnir* association levels of information learning to FRP in semantic learning. The blue lines indicate the prediction process under different routing prefixes in the inference process.

of prefixes, resulting in the construction of an FRP database with broad coverage and high accuracy (§IV-D).

1) Why Autoregressive Model: The FRP pattern is inherently characterized by sequential dependency, where each bit in the prefix depends on preceding bits—a property stemming from IPv6 addressing architecture [34]. Autoregressive models [35] effectively capture this sequential dependence. Moreover, it provides a unified framework for FRP prediction across multiple protocols and port types. As detailed in §III, autoregressive models optimize the alignment between predicted and actual FRP distributions by minimizing KL divergence.

2) Why Transformer Model: The self-attention mechanism in Transformers excels at modeling complex relationships between FRP patterns and influencing factors (e.g., AS numbers, organization types). The multi-head attention mechanism (§IV-B) enables simultaneous focus on semantic features at different positions, which is crucial for learning cross-organizational FRP distribution patterns. Additionally, compared to RNN architectures [36], Transformers offer superior parallel computing capabilities, significantly improving processing efficiency for large-scale IPv6 address spaces.

B. Semantic Learning

As mentioned above, FRP patterns are not randomly distributed—they are influenced by various factors, including protocol and port types, routing prefixes, associated ASes, affiliated organizations, and business categories. These factors exhibit complex semantic relationships that are difficult to capture using simple rule-based approaches or traditional feature engineering. Therefore, we aim to design a highly scalable model to implement a unified FRP prediction algorithm that is protocol- and port-agnostic. At the core of this model is its ability to dynamically generate conditional probability distributions based on the characteristics of different protocols, enabling joint probabilistic modeling of FRP patterns. Therefore, as shown in Figure 3, we adopt a sequence model for semantic learning of FRP patterns. This allows *Gungnir* to leverage self-attention mechanisms and embedding techniques to capture intricate dependencies within the input data. By optimizing cross-entropy loss, the model effectively identifies and avoids FRP Misleading (Misjudging the length of the FRP), ultimately enhancing the accuracy of FRP probing.

Embedding. The embedding layer provides a high-dimensional representation of the input data, enabling the model to capture rich semantic information. To effectively learn the complex relationships among protocol and port types, routing prefixes, belonging ASes, affiliated organizations, and business types, *Gungnir* uses a long sequence for semantic learning. The tokenizer begins by converting the target prefixes into a sequence of tokens, while integrating relevant contextual information, including BGP route announcements, business relationships, and other network attributes. In the tokenization process, the system first assigns sequential identifiers to each token within the sequence information (for example, Cloudflare ID: 124). It then maintains a mapping table that associates these numerical identifiers with their corresponding high-dimensional vector representations, establishing a consistent reference framework for downstream processing. In the process of mapping FRPs to tokens, FRPs are expanded into a binary form representation to ensure that the model can perform more fine-grained learning and prediction. The composition of FRPs is divided into a routing prefix part and an FRP pattern part. These high-dimensional embedding vectors are learnable parameters during semantic learning. To preserve and leverage the sequential nature of the data, positional encoding PE is added to the token embeddings TE, resulting in the final input embeddings IE that are fed into the model. This design enables the model to effectively

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learn sequence-dependent patterns crucial for accurate FRP prediction.

$$IE = TE + PE, \tag{4}$$

To accommodate the variable-length nature of FRP sequences, special tokens—BOS (Begin Of Sentence) and EOS (End Of Sentence)—are added to each sequence to explicitly denote its start and end. By incorporating these boundary markers, Gungnir is able to mitigate the risk of falling into the so-called "FRP Misleading" where incorrect assumptions about sequence boundaries could otherwise compromise prediction accuracy.

Decoder. The decoder consists of multiple encoding layers stacked on top of each other, each consisting of a multi-head self-attention (MHSA) layer, a layer normalization block (LN-block), and a feed-forward neural network (FFN). The encoded embedding vector X is computed with the matrices W_Q, W_K, and W_V, respectively, to obtain:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \tag{5}$$

Matrix operations are performed on the resulting matrices Q, K, V, and attention weights are calculated :

$$\text{Attention} = \text{softmax}\left(\frac{QK^T}{\sqrt{d/h}}\right) \cdot V, \tag{6}$$

where d is the dimension of the key vector and h is the number of attention heads. The results of the computations for each head are stitched together and passed through a linear transformation:

$$\begin{aligned} &\text{MultiHead}(Q, K, V) = \text{Concat}(\text{Attention}_1 \cdots \text{Attention}_h) \cdot W_O \\ &Z = \text{LayerNorm}(X + \text{Dropout}(\text{MultiHead}(Q, K, V))), \end{aligned} \tag{7}$$

where $W_O \in \mathbb{R}^{d \times d}$ is the output projection matrix. To prevent gradient vanishing, residual connections are added to the output X, followed by normalization to obtain the final output Z. The final Z is added to the original input after passing through a feed-forward neural network, and the output is obtained by layer normalization.

Loss Function. During semantic learning, the model can learn the pattern representation of FRPs based on the input sequences in the routing prefix part, which captures the long sequence dependencies. The final loss function optimizes the difference between the predicted probability distribution and the distribution of the true labels by calculating the cross-entropy loss. As mentioned in §III, our core objective is to generate new patterns with high similarity through the original FRP patterns. In this task, the fact that the real FRP pattern distributions are pre-determined stationary distributions allows us to employ a more efficient optimization strategy: minimizing the KL divergence is mathematically exactly equivalent to minimizing the cross-entropy loss.

$$Loss(y, \hat{y}) = -\sum_i y_i \log(\hat{y}_i) \tag{8}$$

where y is the distribution of the true labels, and $\hat{y}_i$ is the distribution of probabilities predicted by the model.

Masking. To have a consistent logical order of sequence generation, all token generation relies only on the presequence tokens that have been generated. We perform a masking operation on the attention scores to ensure that the model learns semantically by associating information such as protocol types and business relationships during semantic learning, and to avoid that during the process of learning FRP's schema, the model can see the future information:

$$\text{Attention}_{i,j} = \text{Attention}_{i,j} + M_{i,j} \tag{9}$$

For an $n \times n$ attention matrix, the mask $M_{i,j}$ can be computed as follows:

$$M_{i,j} = \begin{cases} 0 & \text{if } j \leq i \\ -\infty & \text{if } j > i \end{cases} \tag{10}$$

C. Prefix Inference

In semantic learning, the model has learned FRP patterns under multiple relational attributes, and next, the model needs to generate FRPs for the given conditions. Previous work, such as Luori [25], only predicts single-type information and only for a single active type FRP. To allow the model to generate patterns of FRPs based on joint probabilities given multiple conditions, we employ prefix inference. As shown in Figure 5, its decision process can be viewed as a Markov chain, and for a given known FRP, the model will learn a decision process as follows:

$$\begin{aligned} w_1 &\leftarrow P(w_C \mid w_P) \\ w_2 &\leftarrow P(w_{SC} \mid w_C, w_P) \\ &\cdots \\ w_{i-1} &\leftarrow P(w_{FRP_4} \mid w_{FRP_3} \ldots w_{AS}, w_{ORG}, w_{SC}, w_C, w_P) \\ w_i &\leftarrow P(w_{FRP_5} \mid w_{FRP_4} \ldots w_{AS}, w_{ORG}, w_{SC}, w_C, w_P) \end{aligned} \tag{11}$$

Given the token w, where w_P is the active protocol of the FRP, w_C is the commercial category, w_{SC} is the commercial subcategory, w_{ORG} is the organization to which the FRP belongs, and w_{AS} is the AS to which the FRP belongs.

Sampling. In the process of prefix inference, the model will give the probability value $P(w_i)$ of predicting the next token w for the existing sequences. To balance the *exploitation* and *exploration*, we rank these probabilities $P(w_i)$ from the largest to the smallest and calculate the cumulative probability $C(w_i)$ for the token satisfying the probability p of the token. The tokens that meet the probability p are sampled according to the probability, i.e., Top-p sampling is performed.

$$C(w_i) \geq p \ , C(w_i) = \sum_{j=1}^{i} P(w_j) \tag{12}$$

Therefore, the probability that the next token w_i is selected is

$$P_{next}(w_i) = \frac{w_i}{\sum w} \tag{13}$$

Obviously, the value of p will have an impact on the prediction result, when p is large, the prediction result is more inclined to *exploration*, and when p is small, the prediction result is more inclined to be generated from the data in semantic learning, that is, *exploitation*. We discuss in detail the effect of the value of p in §V-D.

D. Online Scanning

The online scanning module executes probing operations and collects results based on candidate FRP prefixes generated

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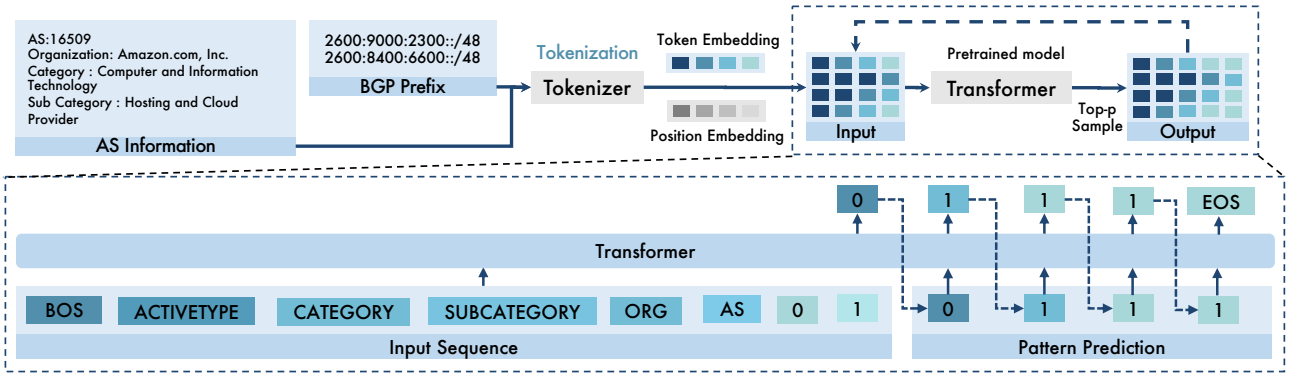


Fig. 5: Prefix inference framework. The top is the overall process by which the model performs prefix inference, and the bottom is the process by which the model specifically makes decisions.

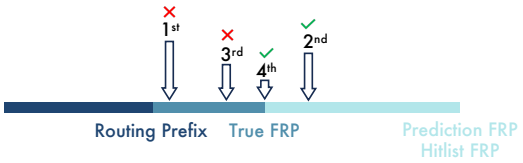


Fig. 6: Granularity correction. After the first probe was performed, no response was obtained, so the probe length was adjusted and the probe was performed again. The above process is repeated until the true FRP particle size is determined.

by the prefix inference module, primarily comprising two stages: FRP confirmation and granularity correction. In the FRP confirmation phase, the module generates IP address probes for each candidate prefix and executes scanning operations. It then analyzes the probing responses to validate FRP prefixes—if all probed IP addresses within a prefix respond affirmatively, the prefix is confirmed as a genuine FRP and incorporated into the FRP database. The subsequent granularity refinement stage adjusts prefix lengths based on probing outcomes, correcting potential granularity inaccuracies in the FRP dataset. This iterative process enhances both the precision of FRP identification and the overall accuracy of subsequent probing operations.

Granularity Correction. As Erdemir et al. [23] found, the current FRP dataset, e.g., the well-known IPv6 hitlist [12], suffers from a large number of prefix granularity errors. For example, a large number of /48 and /56 prefixes are incorrectly categorized as /64 FRPs. The deviation of the prefix length distribution in the dataset from the real web environment where FRPs are deployed affects the performance of the prediction methods, e.g., *Gungnir* for semantic learning learns the wrong sequence distributions, which generates this noise in the prefix inference. To solve this problem, we design the granularity correction mechanism. By correcting the length of existing prefixes, FRPs closer to the true length distribution are obtained. As shown in Figure 6, given the initial correction length l_c, the length of the probed FRPs is gradually taken shorter and l_c is decayed by half in each round.

V. Evaluation

A. Experimental Setting

1) Dataset: We utilized 227,894 IPv6 BGP routing prefixes from the RouteViews project [37] as target prefixes, along with Luori's publicly available FRP dataset [25] as the initial seed set. This dataset contains FRP on multiple ports, including ICMPv6, TCP/22, and TCP/443. Notably, the ICMPv6 class exhibits the highest activity level, comprising 182,942 active FRPs distributed across 4,643 routing prefixes, representing both the largest count and spatial coverage among all categories.

2) Hyper-parameters: The model employs a 6-layer Transformer decoder architecture with 128-dimensional hidden states and 8 attention heads, trained using the Adam optimizer with an initial learning rate of 0.001, batch size of 512, for 50 epochs. During prefix inference, we maintain the same batch size (512) with a sampling probability $p = 0.6$. To ensure fair comparison, all methods probe 16 randomly selected addresses within candidate FRP prefixes, and a prefix is confirmed as an FRP only when all 16 addresses respond positively.

3) Hardware: The experiments were conducted on a host machine equipped with an Intel(R) Xeon(R) Gold 6230R CPU (2.10 GHz), 256 GB of system memory, and a 100 Mbps network connection. For both model training and inference, we utilized a single NVIDIA RTX 4090 GPU with 24 GB of memory.

B. Evaluation Metrics

- Cov_{Space}: This metric quantifies the address space occupied by FRPs while normalizing for prefix length variations. For instance, a /48 prefix and sixteen /52 prefixes each cover 2^{80} addresses ($2^{128-48} = 16 \times 2^{128-52}$), demonstrating equivalent spatial coverage despite differing prefix lengths. By eliminating length-dependent biases, Cov_{Space} provides an unbiased measure of address space utilization. In FRP probing evaluations, discovering more FRPs inherently requires identifying **more and shorter** prefixes—making Cov_{Space} a critical performance metric that directly reflects algorithmic efficiency in capturing fine-grained address blocks.

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TABLE I: Probing results of *Gungnir*, Luori, and MAPD on ICMPv6

B_{prefix}	10			30			50		
$Method$	MAPD	Luori	***Gungnir***	MAPD	Luori	***Gungnir***	MAPD	Luori	***Gungnir***
B_{Total}	2,207,921	2,030,473	1,921,238	6,614,739	5,971,404	6,126,213	11,368,977	9,743,160	10,495,104
$Cov_{Routing}$	660	409	**1158**	809	1202	**1733**	909	1765	**1903**
Cov_{AS}	114	95	**121**	133	129	**156**	143	141	**169**
Cov_{Space}	8.32×10^{22}	2.73×10^{30}	$\mathbf{1.17 \times 10^{31}}$	2.48×10^{23}	4.81×10^{30}	$\mathbf{1.75 \times 10^{31}}$	4.22×10^{23}	5.18×10^{30}	$\mathbf{1.92 \times 10^{31}}$

- Cov_{AS} and $Cov_{Routing}$: AS Coverage (Cov_{AS}) measures the number of unique ASes containing detected FRPs, while Routing Coverage ($Cov_{Routing}$) counts the distinct routing prefixes associated with these FRPs. Together, these metrics evaluate the geographical and topological diversity of detected FRPs. Ideally, an effective probing algorithm should maximize both Cov_{AS} and $Cov_{Routing}$, indicating broader coverage across diverse network infrastructures.
- $Cov_{A \geq B}$: On datasets A and B, the number of routing prefixes under a single routing prefix in the FRP space A is greater than that of B. It is computed as follows: take the overall coverage routing prefix set $R_{_{total}} = R_A \cup R_B$, choose FRPs according to routing prefixes from the set R_{total}, and if a certain routing FRP space A is greater than or equal to B under the prefix, $Cov_{A \geq B}$ is added as 1. This can be expressed as $Cov_{A \geq B} = \mathrm{Bool}\left(\mathrm{S}_{rA} \geq \mathrm{S}_{rB}\right), r \in R_{total}$. The reason for using this metric is that while Cov_{Space} reflects the amount of space occupied by the FRP, this metric is more affected by shorter FRPs.

C. Performance Evaluation

1) Probing Effect: Table I and Table II show the probing results under ICMPv6, TCP/80 type for three different methods: MAPD, Luori, and *Gungnir*. B_{prefix} is the predicted budget under each routing prefix(i.e., the number of probes under a single prefix $b \leq B_{prefix}$), while B_{Total} is the budget used in the actual probing process. We adopt a probing methodology similar to prior work [14], [17], [21], [25], [29], where one budget allows sending probes to 16 addresses within a single prefix. The choice of the three budgets $B_{prefix} = [10, 30, 50]$ is motivated by two factors: (1) prior studies [25] have adopted similar budget values for fair comparison, and (2) these settings enable FRP probing across target prefixes within a manageable and controlled timeframe.

As shown in Table I, *Gungnir* consistently outperforms both MAPD and Luori across all tested prefix budgets ($B_{prefix} = [10, 30, 50]$) in terms of routing prefix coverage ($Cov_{Routing}$) and autonomous system coverage (Cov_{AS}). At $B_{prefix} = 10$, *Gungnir* achieves $Cov_{Routing} = 1158$, increasing the coverage by 2.8 times and 1.7 times compared to Luori and MAPD, respectively. This performance gap further widens at $B_{prefix} = 50$, where *Gungnir*'s $Cov_{Routing} = 1,903$ more than doubles MAPD's result. The coverage space metric (Cov_{Space}) demonstrates even more striking advantages: *Gungnir* maintains a consistent lead of $3.7 - 4.2\times$ over Luori and $4.5 \times 10^7 - 1.4 \times 10^8$ over MAPD. Notably, at $B_{prefix} = 50$, *Gungnir* achieves $Cov_{Space} = 1.92 \times 10^{31}$, surpassing all other methods by significant margins. These results strongly validate the effectiveness of our granularity correction algorithm in precisely determining FRP granularities.

Table II shows the budget and coverage when using different methods for network probing under TCP/80. Again, *Gungnir* shows significant advantages at all budgets. At $B_{prefix} = 10$, $Cov_{Routing}$ of *Gungnir* is 574, which is 3.4 times that of Luori and 1.9 times that of MAPD. When B_{prefix} increases by 50, $Cov_{Routing}$ of *Gungnir* reaches 981, which is 1.5 times that of Luori. Meanwhile, the Cov_{Space} of *Gungnir* stays maximal under all budgets, which is 2.4-3.0 times higher than that of Luori, while that of MAPD ranges from $1.54 \times 10^{22} - 2.12 \times 10^{30}$ times.

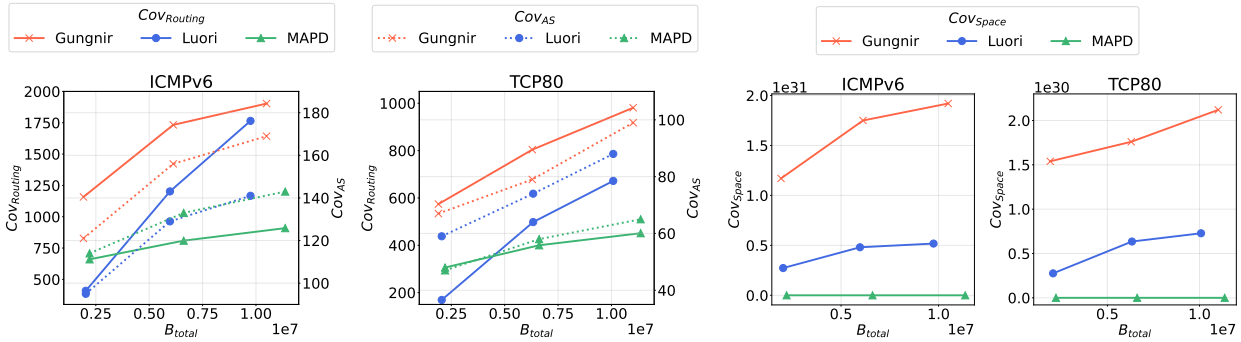
Due to differing budget structures among the three methods, we employ a Pareto chart (Figure 7) for performance comparison, with the y-axis representing actual budget usage (B_{Total}). Figures 7a and 7b present results for ICMPv6 and TCP/80 across $Cov_{Routing}$/Cov_{AS} and Cov_{Space} metrics respectively. Superior methods are indicated by curves closer to the upper-left corner, denoting greater coverage per budget. *Gungnir* demonstrates consistent dominance across all metrics, maintaining strong performance even at low budgets ($B_{prefix} = 10$) through its flexible length prediction strategy. In contrast, Luori's layer-by-layer search tree approach underperforms MAPD at minimal budgets due to its rigid expansion mechanism.

2) Operating Time: Table III compares the performance of both Luori and *Gungnir* in terms of operating time. The experiment assigns 50 budgets to each routing prefix across the Internet, and the average of multiple experiments is taken to count the running time. The pre-training time and inference time counted here refer to the total time, and N refers to the number of target protocols and ports (e.g., TCP/80). Since Luori can only build a tree for a single active type, Luori's method takes about $4.9 \times N$ hours for pre-training, while *Gungnir* uses global information for semantic learning, so *Gungnir* takes only 6.8 hours in total, and the global information used also contributes to the accuracy and comprehensiveness of the predictions. In the inference phase, Luori's method takes about $85 \times N$ hours, while the *Gungnir* method takes only about $2.5 \times N$ hours, achieving $\times 34$ increase in inference times on just a single active type. This means that if the FRP is probed for 5 active types, *Gungnir* can achieve nearly 7 times improvement in pre-training time compared to Luori, and more than 27 times improvement in total execution time. This shows the higher efficiency and more comprehensive probing capability of *Gungnir*.

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TABLE II: Probing results of *Gungnir*, Luori, and MAPD on TCP/80

B_{prefix}	10			30			50		
$Method$	MAPD	Luori	***Gungnir***	MAPD	Luori	***Gungnir***	MAPD	Luori	***Gungnir***
B_{Total}	2,207,921	2,051,384	1,897,063	6,614,739	6,334,095	6,293,054	11,368,977	10,083,623	11,016,352
$Cov_{Routing}$	306	169	**574**	400	498	**804**	451	672	**981**
Cov_{AS}	47	59	**67**	58	74	**79**	65	88	**99**
Cov_{Space}	2.88×10^{22}	2.76×10^{29}	$\mathbf{1.54 \times 10^{30}}$	8.54×10^{22}	6.35×10^{29}	$\mathbf{1.76 \times 10^{30}}$	1.44×10^{23}	7.28×10^{29}	$\mathbf{2.12 \times 10^{30}}$

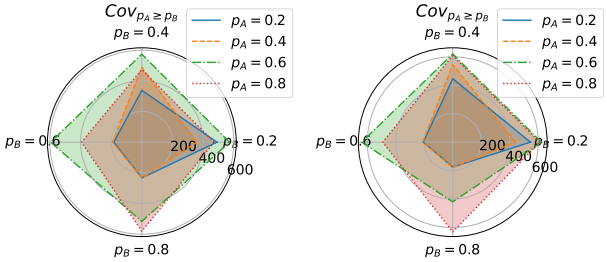


(a) $Cov_{Routing}$ and Cov_{AS} on ICMPv6 and TCP/80 (b) Cov_{Space} on ICMPv6 and TCP/80

Fig. 7: Performance comparison of *Gungnir*, Luori, and MAPD

TABLE III: Comparison of Luori and *Gungnir* on operating time. Luori[1] can only predict routing prefixes that have a seeded matchable relationship under a single protocol.

Method		Luori	*Gungnir*
Pre-training time (h)		$\approx 4.9 \times N$	**6.8**
Inference time (h)		$\approx 85 \times N$	$\approx \mathbf{2.5 \times N}$
Probing range		Most[1]	**All**
Global information		×	✓
Total operating time (h)	$N = 1$	89.9	**9.3**
	$N = 3$	269.7	**14.3**
	$N = 5$	449.5	**19.3**



(a) w/o Granularity Correction (b) w/ Granularity Correction

Fig. 8: Comparison of FRP set relativity metric $Cov_{p_A \geq p_B}$ under different hyper-parameters

D. Value of the Hyperparameter p

The sampling probability p serves as a crucial parameter in *Gungnir*'s prediction mechanism, controlling the trade-off between pattern exploration and exploitation. Lower values of p (approaching 0) cause *Gungnir* to predominantly reproduce patterns similar to those in the original dataset, while higher values (approaching 1) drive the system to predict novel patterns. To systematically evaluate this parameter's impact, we conducted experiments using uniformly sampled values of p across the [0,1] interval, specifically testing at $p \in \{0.2, 0.4, 0.6, 0.8\}$. Since the ICMPv6 tends to be the most active in the probing of FRP, we selected it as our baseline protocol and performed all predictions within BGP-announced routing prefixes. Each routing prefix was predicted 10 times in our experiments, with FRPs subsequently filtered out from the results. The experimental results for different hyperparameter values of p are presented in Table IV. When p is set to 0.2 or 0.4, the model demonstrates inferior performance on both Cov_{AS} and $Cov_{Routing}$ metrics compared to values of 0.6 and 0.8, as these lower values constrain the model to generate FRPs primarily within already learned patterns. Notably, p values of 0.6 and 0.8 yield comparable results for both coverage metrics.

To conduct a more comprehensive comparison, we employed the relative metric $Cov_{p_A \geq p_B}$, which evaluates the spatial sizes of results across different routing prefixes while mitigating the influence of excessively short FRPs on spatial measurements. Figure 8 illustrates this comparison, where the axis represents set B and the four lines correspond to set A with p_A values of $\{0.2, 0.4, 0.6, 0.8\}$. The analysis reveals that $Cov_{p_{0.6} \geq p_B}$ for $p_A = 0.6$ generally exceeds $Cov_{p_{0.8} \geq p_B}$ for $p_A = 0.8$, except when $p_B = 0.8$. This exception occurs because the metric inherently favors cases where sets A and B are identical. Importantly, in all other comparisons, $p_A = 0.6$ demonstrates superior performance to $p_A = 0.8$, both before and after granularity correction. These findings suggest that while $p = 0.8$ produces larger FRP spaces, these gains are likely attributable to a limited number of shorter FRPs. This interpretation is supported by Table IV, which shows that $p = 0.8$ fails to significantly improve Cov_{AS} and $Cov_{Routing}$ metrics and underperforms $p = 0.6$ in some cases. Consequently, we recommend using $p = 0.6$ as the optimal sampling value for prefix inference tasks.

E. Effectiveness of Granularity Correction

To verify that *Gungnir*'s granularity correction can effectively avoid falling into the "FRP Misleading", we count the

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TABLE IV: Ablation study of granularity correction

p	0.2	0.4	0.6	0.8
Before Granularity Correction				
$Cov_{Routing}$	490	488	**595**	581
Cov_{AS}	90	91	**110**	108
Cov_{Space}	3.67×10^{26}	3.61×10^{26}	5.06×10^{26}	$\mathbf{5.29 \times 10^{26}}$
After Granularity Correction				
$Cov_{Routing}$	546	544	**640**	631
Cov_{AS}	92	93	**113**	112
Cov_{Space}	5.34×10^{30}	5.52×10^{30}	1.13×10^{31}	$\mathbf{1.66 \times 10^{31}}$

cases before and after applying the mechanism under these four sets of parameters. From Table IV, we can see that the experimental results with granularity correction have a certain amount of improvement in $Cov_{Routing}$ and Cov_{AS}, while the results with granularity correction are significantly higher than the ones without granularity correction in Cov_{Space}, which is about $\times 10^5$ times higher in the space. This is consistent with our motivation for designing the granularity correction, i.e., to avoid the original dataset and the prediction process from causing *Gungnir* to fall into the *FRP Misleading*.

F. Long-term Multi-protocol Probing

Through continuous Internet-wide monitoring spanning over three months, we have compiled the most comprehensive FRP dataset to date. Our collection encompasses 23 distinct active protocol types, including ICMPv6, TCP, and UDP, representing the broadest coverage and largest address space currently available. As demonstrated in Table V, comparative analysis reveals that our dataset significantly outperforms existing FRP lists across three key metrics: (1) coverage space, (2) number of covered routing prefixes, and (3) count of included ASes, with ICMPv6 serving as the primary benchmark for these comparisons.

G. FRP Deployment

We conduct a comprehensive analysis of the Internet-wide FRP distribution across multiple protocols, employing both quantitative and spatial metrics to evaluate deployment scales. As detailed in Appendix Tables VII and IX, the distribution patterns reveal significant variations across major protocols. Notably, three major organizations—Cloudflare, Amazon, and Akamai—dominate the ICMPv6 FRP count, collectively accounting for over 2 million FRPs. Their deployment strategies contrast sharply with organizations like SIXNET: while the latter occupies vast address spaces, Cloudflare and Amazon implement fine-grained subnet segmentation, configuring dense FRP distributions within smaller prefixes. This aligns with known implementations like Linux's *net.ipv6.ip nonlocal bind* option [25], which enables single-host gateway deployments with prefix-wide address binding. Currently, we have identified these deployed FRPs within at least 9 ASes by email and have not received any error feedback from any of the ASes, indicating that our measurements are true and reliable.

H. Active Type Fingerprint

To identify FRPs, we conducted systematic scans of all BGP-advertised routing prefixes across the Internet, focusing on ICMPv6 responsiveness and common TCP/UDP ports. The scanning methodology targeted a comprehensive set of network protocols, with detailed port mappings and their associated system functions documented in Appendix Table VI. Through this large-scale evaluation, we successfully collected and categorized 47 distinct active protocol fingerprints based on their responsiveness characteristics. These fingerprints, representing diverse network behaviors and service configurations, are fully enumerated in Appendix Table X. The fingerprint classification scheme accounts for both protocol-level responsiveness patterns and port-specific behavioral signatures.

TABLE V: Comparison of our ***Our FRPlist***, Luori's FRPlist, and Hitlist's FRPlist

Dataset	Publication time	Cov_{Space}	Cov_{AS}	$Cov_{Routing}$
Our FRPlist	April 30, 2025	$\mathbf{2.45 \times 10^{32}}$	**624**	**8734**
Luori FRPlist	April 30, 2025	1.95×10^{32}	530	7324
Hitlist's FRPlist	April 30, 2025	1.01×10^{30}	335	3237

Our analysis of active port fingerprints across Internet-wide routing prefixes reveals significant variations in security policies among service providers. Cloud providers like Akamai, Bharti, and Google (2,324 routing prefixes) employ a minimal exposure strategy, permitting only ICMPv6 for basic network connectivity while keeping other ports closed. In contrast, major web service providers, including Amazon, Cloudflare, and Fastly (1,833 routing prefixes), maintain a more open configuration, enabling both ICMPv6 and essential web ports (TCP/80 for HTTP service discovery and TCP/443 for SSL/TLS encryption). A smaller but notable group of hosting providers, such as Rohmad and IP SERVER (395 routing prefixes), exhibit a distinct pattern by opening TCP/22 alongside ICMPv6, suggesting large-scale SSH service offerings. The UDP landscape shows particularly selective deployments, with only a few autonomous systems like NAMESHIELD, Fastly, and Alibaba maintaining FRPs on UDP/53, likely supporting external DNS resolution services with potential load-balancing capabilities across multiple IP addresses. These findings demonstrate how different service models and operational requirements shape distinct network security postures across the IPv6 ecosystem.

I. FRP Service Analysis

To understand why organizations widely deploy FRPs with open ports, we analyzed services under 102,314 prefixes. Using Zgrab2 [38], we probed 1.6 million generated addresses to identify active services. Appendix Table VIII shows service distribution patterns across major providers, revealing their operational needs behind FRP configurations.

1) HTTP: We observe that certain ISPs (e.g., Internet Rimon and Hughes) and hosting providers (e.g., WOODSNET) expose port 80 but do not provide HTTP services. We know from surveys of email and other methods that many organizations will use the addresses under FRP as an address pool, and at the egress gateway, the address pool will assign a proxy ip for the egress service as a random proxy ip for

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the outbound service, and in this way can greatly alleviate the problem of a single IP limiting [39], [40] the speed of the service. This may explain why there is such a huge address space open for specific ports, but no corresponding service configuration. However, this FRP rate-limiting bypass strategy also means that the traditional single-ip rate-limiting and DDoS defense strategy is invalidated, and the traditional IPv4-based security protection mechanisms (such as ACL rules) may not be able to effectively identify the abnormal traffic of IPv6 fully responsive prefixes, which may generate security risks. It is important to note that if these hosts only open ports without actually providing any services, this port exposure to the public network is likely to be a security risk.

Among the generated addresses, $233,198(15\%)$ were successfully identified as hosting HTTP services. These responsive addresses spanned 2,345 routing prefixes managed by 183 distinct organizations with confirmed HTTP service deployments. Notably, Akamai, Amazon, and Incapsula emerged as the top three providers—a finding consistent with our prior analysis. This alignment reinforces our observation that CDNs frequently implement FRPs by enabling HTTP (port 80) across all addresses within a prefix. Such configurations serve as a critical mechanism for these networks to efficiently deliver outbound services at scale.

2) SSH: Among the SSH services scanned, 48,120 generated addresses were successfully identified, spanning 573 routing prefixes across 184 autonomous systems (ASes). Our analysis revealed several security risks, including instances running outdated versions (e.g., OpenSSH 7.2p2) that retain support for deprecated cryptographic algorithms such as diffie-hellman-group14-sha1 [41]. A significant portion of these FRPs likely belongs to a honeypot system designed to dynamically map unused IPv6 addresses to trap servers for attack monitoring and analysis. In this setup, when an attacker scans a network segment, unregistered address requests are redirected to a honeypot, which consolidates multiple trap addresses into a single service IP. This architecture enables efficient aggregation of attack traffic across a vast address space—explaining the observed concentration of similarly configured hosts under shared prefixes. The deliberate retention of OpenSSH 7.2p2 may serve as an active lure, exploiting attackers' propensity to target known vulnerabilities (e.g., CVE-2017-15906). By maintaining compatibility with weak algorithms (e.g., diffie-hellman-group14-sha1), the honeypot could further act as a protocol-level decoy, capitalizing on scanning tools' tendency to prioritize such algorithms to enhance detection rates.

Additionally, a small number of non-honeypot hosts were identified. To enhance security, we recommend enforcing key-based authentication (disabling passwords), changing the default SSH port, restricting access via firewall rules, and deploying tools like Fail2ban to block brute-force attacks.

3) SSL/TLS: Our analysis identified 20,391 SSL/TLS-enabled addresses across 579 routing prefixes and 113 ASes. A significant portion of FRPs utilize shared certificates, particularly from Cloudflare (e.g., Cloudflare issues certificates for its domain cdnjs.cloudflare.com and massively deploys the certificates under its FRP) and other providers employing wildcard certificates (*.xxxx.com). While this approach streamlines HTTPS content delivery and reduces certificate management costs, it introduces potential risks - a single compromised private key could affect multiple nodes. Certificate validity analysis shows most vendors comply with industry standards for short validity periods. However, for the few with extended validity, we recommend implementing automated certificate management systems for FRPs. Additional details are available in Appendix Figure 9b.

VI. Ethical Considerations

In order to avoid impacting the real Internet [42], [43], all our experiments are carried out at a single measurement point using a single host, and the scanning speed is controlled at 1 Mbps. With different BGP routing prefixes, our scanning requests are dispersed at different times during the test time period, and the destination addresses are completely broken up, and at the same time, a single scanning request is not more than 10 minutes. The experiments provide an exit mechanism, and for ASes that do not wish to be detected by the scan, we include them in the blocking table and do not conduct the experiments. In order to avoid attacks on real organizations and network infrastructures, for possible vulnerabilities, we only judged by fingerprint discrimination and version information, and did not perform any penetration tests on real hosts, etc., to verify the existence of vulnerabilities.

VII. Conclusion

In this paper, we propose an autoregressive model-based multi-protocol unified FRP probing algorithm named *Gungnir*, which aims to address the network asset scanning challenges posed by FRPs in IPv6 networks. *Gungnir* uses autoregressive modeling to learn FRP pattern representations through semantics by capturing the relationships among FRPs. With the same budget, *Gungnir* improves the probing efficiency by nearly 27 times, extends the coverage space by 4.2 times, and covers more autonomous systems and routing prefixes to achieve SOTA performance. Additional experiments validate the effectiveness of the hyperparameter and granularity correction mechanisms. To facilitate better community research in active addresses and FRPs, this paper makes the code and all data publicly available and constructs a list of the most extensive FRPs containing multiple active types. Furthermore, it provides new ideas for Internet asset scanning and network security analysis.

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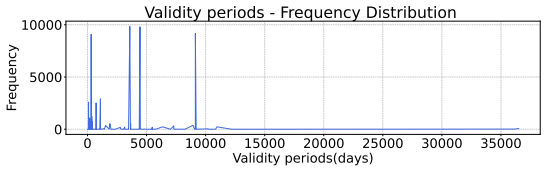
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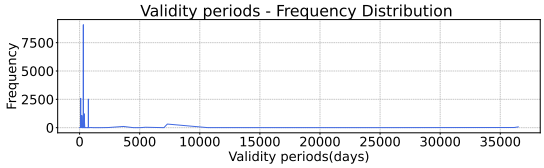
APPENDIX

TABLE VI: Common ports and their corresponding services

Port	TCP/21	TCP/22	TCP/25	TCP/80	TCP/110	TCP/443	TCP/1433	TCP/3306	UDP/53
Service	FTP	SSH	SMTP	HTTP	POP3	HTTPS	SQL Server	MySQL	DNS



(a) Distribution of validity period of address certificates under FRP



(b) Distribution of validity period of address leaf certificates under FRP

Fig. 9: Distribution of validity period of address all/leaf certificates under FRP

TABLE VII: Top 10 ASes with active FRPs for ICMPv6, TCP/80, and TCP/443

ICMPv6			TCP/80			TCP/443		
AS	ORG	#FRPs	AS	ORG	#FRPs	AS	ORG	#FRPs
13335	Cloudflare	1,010,784	20940	Akamai	321,472	6621	Hughes	194,496
16509	Amazon	661,984	6621	Hughes	200,192	20940	Akamai	42,784
20940	Akamai	447,872	19551	Incapsula	170,036	16509	Amazon	22,688
36183	Akamai	241,376	55547	WoodsDale	52,224	22773	Cox	19,840
32934	Facebook	159,872	47583	Hostinger	32,720	9009	M247	14,240
19551	Incapsula	57,216	42925	Internet Rimon	27,616	62240	Clouvider	12,800
47583	Hostinger	35,321	16509	Amazon	22,368	42003	OGERO	12,128
34984	Superonline	28,128	215304	YUWAN	20,384	40509	Fly.io	11,200
204916	RACKTECH	28,032	22773	Cox	13,440	28613	HUGHES	10,304
36492	Google	27,936	10881	FUNPAR	9,728	197207	mci.ir	9,600

TABLE VIII: Top 10 ASes with hosts offering HTTP, SSL/TLS, and SSH services

HTTP			SSL/TLS			SSH		
AS	ORG	Counts	AS	ORG	Counts	AS	ORG	Counts
20940	Akamai	4,368,512	47583	Hostinger	305,344	214610	YangXu Liao	115,456
16509	Amazon	663,072	34762	Combell NV	81,088	62240	Clouvider	87,200
19551	Incapsula	648,416	19551	Incapsula	50,464	60781	LeaseWeb Netherlands	86,112
47583	Hostinger	302,912	13335	Cloudflare	23,712	51396	Pfcloud UG	80,192
13335	Cloudflare	269,184	20940	Akamai	23,584	61400	Start LLC	63,840
7713	PT Telekomunikasi	82,336	198066	Grupo Loading Systems	20,352	214040	RUNNAN WEN	59,296
34762	Combell NV	82,272	210403	Groupe LWS SARL	18,688	44812	IP SERVER LLC	58,144
9500	One New Zealand Group	78,304	12654	RIPE NCC	17,760	44354	BROCK VOJKOVIC	55,968
54113	Fastly	75,232	48254	20i Limited	11,488	204490	Kontel LLC	52,416
61400	Start LLC	66,560	216271	NETLABS LLC	10,624	215304	YUWAN NETWORKS	48,224
215575	HIGH PING	42,048	63410	PrivateSystems	8,096	6939	Hurricane Electric LLC	42,368
6762	TELECOM ITALIA	36,416	200260	Debugged Ltd	7,904	214911	Li Hang	37,120

TABLE IX: Top 10 ASes by Cov_{Space} of active FRPs collected for ICMPv6, TCP/80, and TCP/443

ICMPv6			TCP/80			TCP/443		
AS	ORG	Space	AS	ORG	Space	AS	ORG	Space
214168	SIXNET	2.22×10^{31}	51088	A2B	3.80×10^{30}	51088	A2B	3.80×10^{30}
60781	LeaseWeb	2.08×10^{31}	18779	EGIHosting	3.80×10^{30}	216271	NETLABS	6.33×10^{29}
215158	Peerlink LTD	1.39×10^{31}	204916	RACKTECH	1.59×10^{30}	205807	Essener	6.33×10^{29}
34984	Superonline	1.15×10^{31}	61400	Start	7.92×10^{29}	209242	Cloudflare	2.37×10^{29}
216157	SIXNET	8.95×10^{30}	210644	AEZA	6.34×10^{29}	42692	Overweb	1.58×10^{29}
197992	DEFSOLUTION	7.77×10^{30}	33856	Turkiye	6.34×10^{29}	54113	Fastly	7.92×10^{28}
28753	Leaseweb	7.29×10^{30}	216271	NETLABS	6.34×10^{29}	3462	hinet	7.92×10^{28}
46573	LayerHost	6.97×10^{30}	43350	NForce	6.34×10^{29}	34939	nextdns	7.92×10^{28}
215427	revonia	6.34×10^{30}	205807	Essener	6.34×10^{29}	202888	UPSTREAM	7.92×10^{28}
18779	EGIHOSTING	5.70×10^{30}	34665	Petersburg	3.91×10^{29}	214040	RUNNAN WEN	2.47×10^{28}

TABLE X: FRP fingerprint for 227,894 BGP routing prefixes across the entire Internet

ICMPv6	TCP								UDP	Counts	Typical Type
	21	22	25	80	110	443	3306	1433	53		
										222,572	
✓										2,324	Akamai/Bharti/Google
✓				✓		✓				1,833	Amazon/Cloudflare
✓		✓								395	Rohmad/IP SERVER
✓	✓		✓	✓	✓	✓	✓	✓		220	Incapsula
✓	✓	✓	✓	✓	✓	✓	✓	✓		105	Fly.io/Cloudflare
		✓								100	Vostrikov/Orange Tunisie
✓				✓						54	Amazon/Fastly
✓		✓		✓		✓				52	RADISHCLOUD/YUWAN
						✓				42	M247/Amazon
✓		✓		✓						29	Start/Pfcloud
				✓		✓				22	Amazon/A2B
	✓	✓	✓	✓	✓	✓	✓	✓		22	LIBANTELECOM/A2B
	✓	✓	✓	✓	✓	✓	✓	✓		19	Cloudflare/Fastly
	✓		✓	✓	✓	✓	✓	✓		13	Vostrikov
			✓		✓					12	CERNET
				✓						12	CERNET
✓		✓				✓				7	YUWAN NETWORKS
✓			✓	✓						6	EGIHosting
✓		✓					✓			5	Nanning Huifa
	✓	✓	✓	✓	✓		✓	✓		5	LIBANTELECOM
					✓					3	CERNET
✓									✓	3	NAMESHIELD/Fastly
✓	✓	✓	✓	✓	✓		✓	✓		3	Alibaba
✓	✓	✓		✓	✓	✓	✓	✓		3	Fly.io
		✓		✓						3	YUWAN NETWORKS
✓		✓		✓			✓			3	Mingyang Shen
			✓	✓		✓				2	GOVERNMENT
✓							✓	✓		2	Sneaker
									✓	2	Alibaba
✓		✓			✓					2	Cr0bar
✓		✓	✓	✓		✓	✓			2	Jikai Ren
✓	✓	✓		✓	✓		✓	✓		2	YUWAN NETWORKS
✓		✓		✓		✓	✓			2	UPSTREAM
			✓							1	Kontel
	✓	✓								1	Turbo Technology
✓	✓	✓	✓	✓		✓	✓	✓		1	NeocomISP
	✓		✓	✓	✓	✓				1	Chunghwa Telecom
	✓			✓		✓				1	Iron Mountain
	✓		✓	✓	✓					1	Cloudflare
✓	✓		✓	✓	✓		✓	✓		1	Incapsula
✓	✓	✓		✓		✓				1	XuanMing
✓	✓	✓	✓	✓			✓	✓		1	Skhron OU
✓	✓	✓	✓		✓					1	Sky Telecom
✓		✓	✓	✓	✓	✓				1	Equinox
							✓			1	RACKTECH
		✓		✓		✓				1	Alexandre

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